

EXPERIMENT 19: Prolog Program for STUDENT-TEACHER-SUB-CODE**Aim**

To write and execute a Prolog program to represent the relationship between students, teachers, and subject codes and retrieve the required information using queries.

Algorithm

1. Start the Prolog program.
2. Define facts representing students, teachers, and subject codes.
3. Establish relationships between students, teachers, and subjects.
4. Define rules or queries to retrieve related information.
5. Execute the required queries.
6. Display the corresponding student, teacher, or subject information.
7. Stop.

Program Code:

% Student database

student(sameer, cse).

student(rahul, cse).

student(priya, aiml).

% Teacher database

teacher(dr_kumar, ai).

teacher(dr_priya, dbms).

teacher(dr_raj, c).

% Subject database

subject(ai, dr_kumar).

subject(dbms, dr_priya).

subject(c, dr_raj).

% Find teacher for a subject

teacher_of(Subject, Teacher) :-

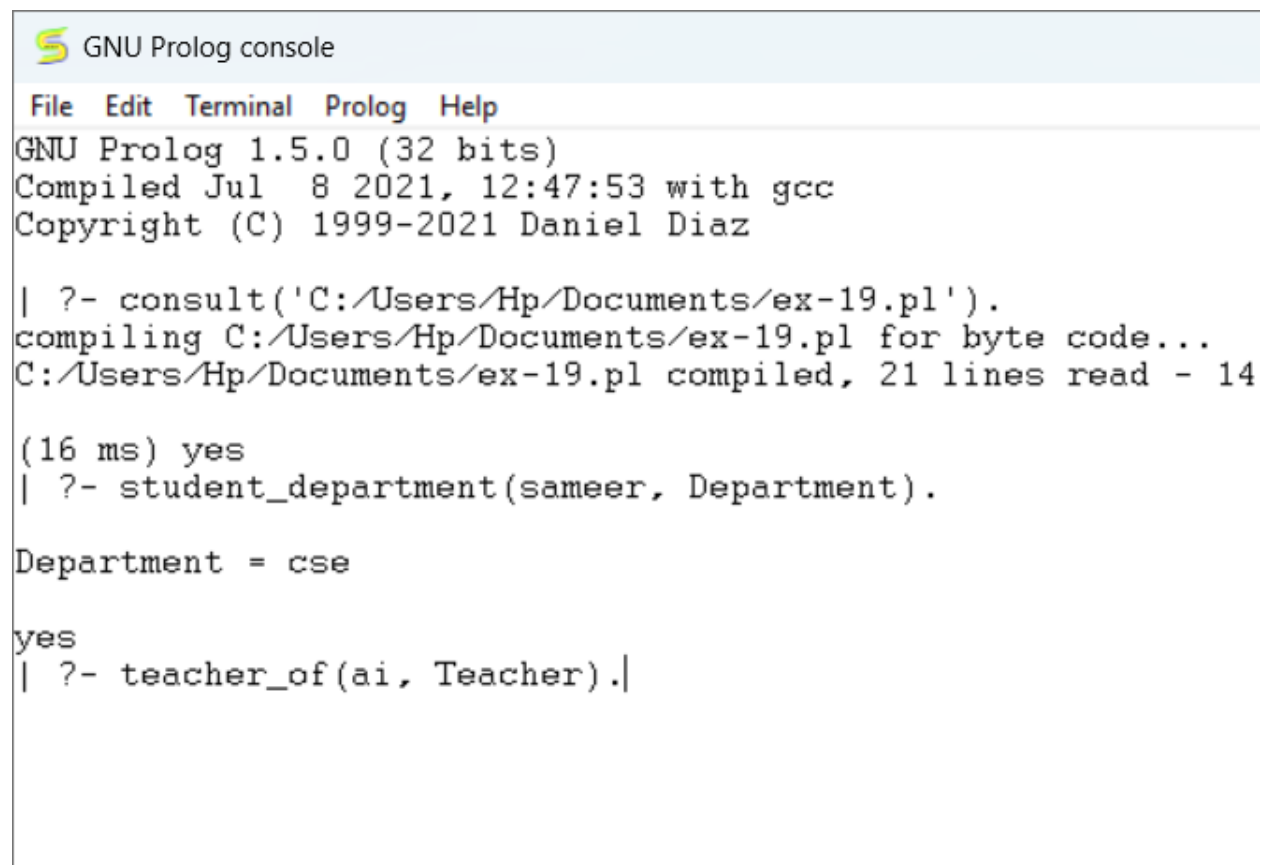
subject(Subject, Teacher).

% Find student's department

student_department(Student, Department) :-

student(Student, Department).

Output:

A screenshot of the GNU Prolog console interface. The title bar says "GNU Prolog console". The menu bar includes "File", "Edit", "Terminal", "Prolog", and "Help". The main text area shows the following output:

```
GNU Prolog 1.5.0 (32 bits)
Compiled Jul  8 2021, 12:47:53 with gcc
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| ?- consult('C:/Users/Hp/Documents/ex-19.pl').
compiling C:/Users/Hp/Documents/ex-19.pl for byte code...
C:/Users/Hp/Documents/ex-19.pl compiled, 21 lines read - 14

(16 ms) yes
| ?- student_department(sameer, Department).

Department = cse

yes
| ?- teacher_of(ai, Teacher).|
```

Result

The Prolog program representing the student-teacher-subject code relationships was successfully implemented, and the required relationships were retrieved correctly using queries.